

Mail Customer Segmentation

Exploratory Data Analysis Report

Internship Task 1

Date: May 23, 2026

Tool Used: Python (Pandas, Matplotlib) | Google Colab

Submitted By:

Bhumi Kumari

1. Introduction

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Mall Customer Segmentation dataset. The primary objective was to understand customer demographics, income distribution, and spending behavior to derive actionable business insights.

The dataset contains information about 200 mall customers, capturing key attributes such as gender, age, annual income, and a spending score assigned by the mall. The analysis was performed using Python libraries — Pandas for data manipulation and Matplotlib for data visualization — in a Google Colab environment.

2. Dataset Overview

2.1 Basic Information

The dataset (Mall_Customers.csv) consists of 200 rows and 5 columns. Below is a summary of the dataset structure:

#	Column Name	Non-Null Count	Dtype
0	CustomerID	200	int64
1	Gender	200	object
2	Age	200	int64
3	Annual Income (k\$)	200	int64
4	Spending Score (1-100)	200	int64

Key observations from the dataset structure:

- The dataset contains 200 entries with no missing values in any column.
- Four columns are of integer type (int64) and one column (Gender) is of object type.
- CustomerID serves as a unique identifier and is not used for analytical purposes.

2.2 Summary Statistics

The table below summarizes the descriptive statistics for all numerical columns:

Column	Count	Mean	Min	Max
CustomerID	200	100.5	1	200
Age	200	38.85	18	70
Annual Income (k\$)	200	60.56	15	137

Spending Score (1-100)	200	50.20	1	99
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2.3 Missing Values

A missing value check was performed on all columns. The result confirmed that the dataset is complete — no null or missing values were found across any of the 5 columns. This indicates a clean, ready-to-analyze dataset.

3. Exploratory Data Analysis

3.1 Age Distribution

A histogram was plotted to understand the distribution of customer ages across the mall dataset. Key findings from the age distribution:

- The customer age ranges from 18 to 70 years.
- The average age of customers is 38.85 years, indicating a predominantly working-age customer base.
- The distribution shows a higher concentration of younger customers in the 20–40 age bracket.
- The 25th percentile age is 28.75 years and the 75th percentile is 49 years.
- Fewer customers are aged above 60 years, suggesting the mall primarily attracts younger to middle-aged shoppers.

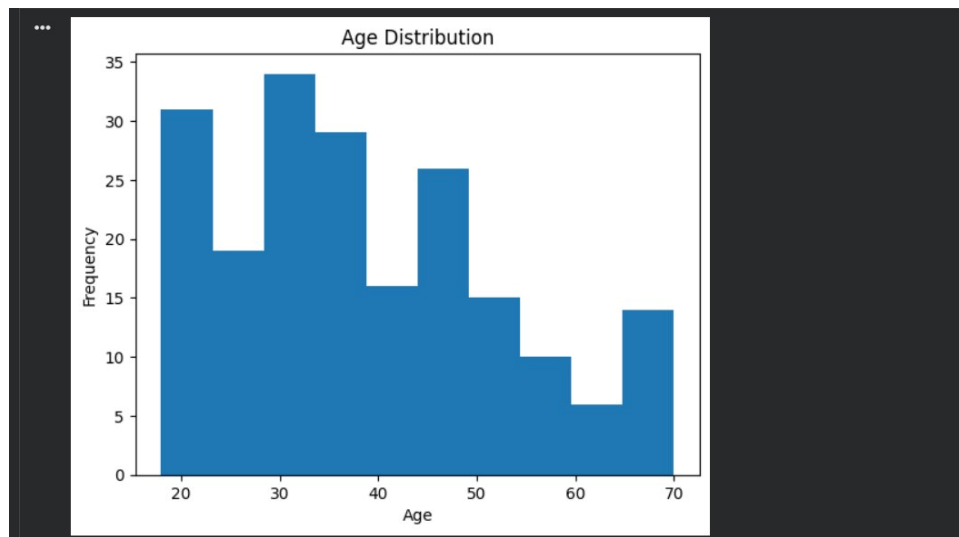


Figure 1: Age Distribution Histogram — Most customers fall in the 20–40 age range

3.2 Gender Distribution

A pie chart was generated to visualize the gender composition of mall customers. The distribution revealed the following:

- Female customers constitute 56% of the total customer base.
- Male customers make up 44% of the total customer base.
- The slight female majority may influence product stocking and marketing strategies.

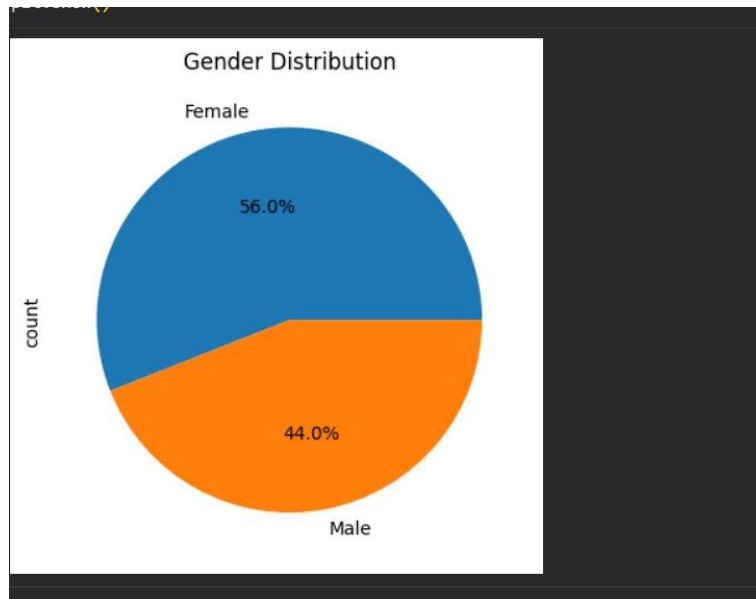


Figure 2: Gender Distribution Pie Chart — 56% Female vs 44% Male

3.3 Average Spending Score by Gender

A bar chart was used to compare the average spending scores between male and female customers. The analysis showed:

- Female customers demonstrated a slightly higher average spending score (~51.5) compared to male customers (~48.5).
- Both genders had broadly similar purchasing behavior with scores in the moderate range.
- The difference in scores indicates female customers may be slightly more engaged with mall offerings.

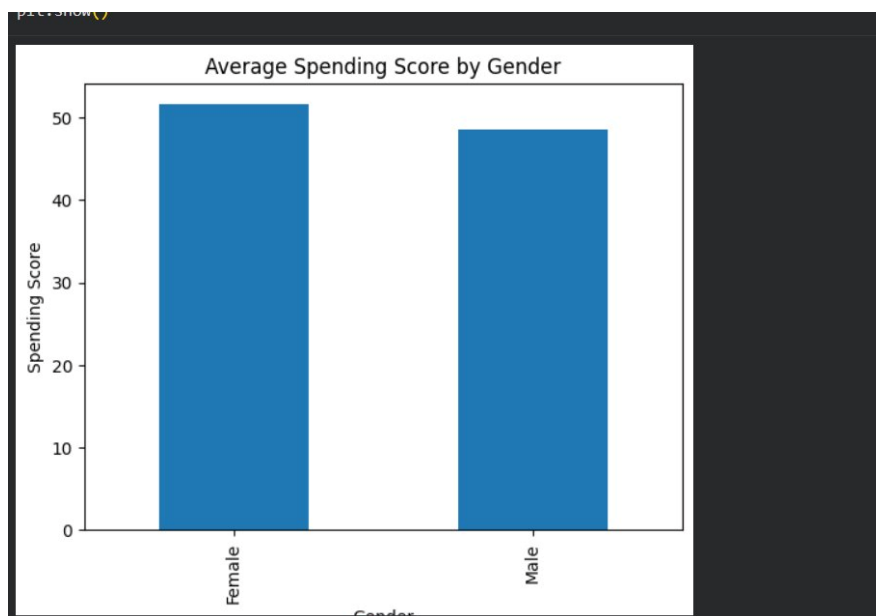


Figure 3: Average Spending Score by Gender — Females score marginally higher

4. Key Findings Summary

The following table consolidates the major findings from the exploratory analysis:

Finding	Value	Insight
Female Customers	56%	Majority demographic
Male Customers	44%	Significant minority
Avg. Age	38.85 years	Working-age adults
Avg. Income	\$60,560/yr	Middle-income segment
Avg. Spending Score	50.20 / 100	Moderate engagement
Missing Values	0	Clean dataset

5. Limitations

The following limitations were identified during the analysis:

- The dataset contains only 200 customer records, which may not fully represent all customer demographics.
- Several behavioral factors — such as purchase frequency, preferred product categories, and seasonal patterns — were not available.
- The analysis relies on historical snapshot data and does not account for temporal trends.
- The Spending Score's exact calculation methodology is not disclosed, limiting its interpretability.

6. Recommendations

Based on the findings of this analysis, the following recommendations are proposed:

- Design targeted and personalized marketing campaigns for younger customers (aged 20–35), as this segment represents the largest portion of the base.
- Customer engagement strategies — loyalty programs, discount offers, exclusive deals — should prioritize high-spending groups to maximize revenue.
- Given that female customers make up 56% and exhibit higher spending scores, promotional activities tailored to female preferences could yield better returns.
- Additional behavioral data — visit frequency, basket size, product category preferences — should be collected for deeper customer profiling.
- Future analyses should include clustering techniques (e.g., K-Means) to enable data-driven customer segmentation.

7. Conclusion

This exploratory analysis of the Mall Customer Segmentation dataset successfully uncovered key patterns in customer demographics and spending behavior. The dataset was found to be clean and complete, making it suitable for further advanced analysis.

The findings indicate that the customer base is predominantly young to middle-aged, with a slight female majority. Spending behaviors are broadly moderate across genders, with female customers showing marginally higher engagement. These insights can serve as a foundation for targeted marketing, customer retention strategies, and future segmentation modeling efforts.